Source Code:

```
 package servlet;  import java.io.IOException;
import javax.servlet.ServletException;  import
javax.servlet.annotation.WebServlet;  import
javax.servlet.http.HttpServlet;  import
javax.servlet.http.HttpServletRequest;  import
javax.servlet.http.HttpServletResponse;
import java.io.*;
@WebServlet("/WebApplication1")  public
class MyServlet extends HttpServlet { private
static final long serialVersionUID = 1L;
public MyServlet() {
  super();
    }
   protected void doGet (HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html");        PrintWriter out = response.getWriter();
out.print("<html><head>My First Servlet</head><body>");
out.print("<center><h2>");
 out.print("Welcome to HTTP Servlets</h2></center>");         out.print("<p>doGet()
Method is executing....</p>");
out.print("</body></html>");
    }
}
```

Output:

RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) HTML Form (index.html):

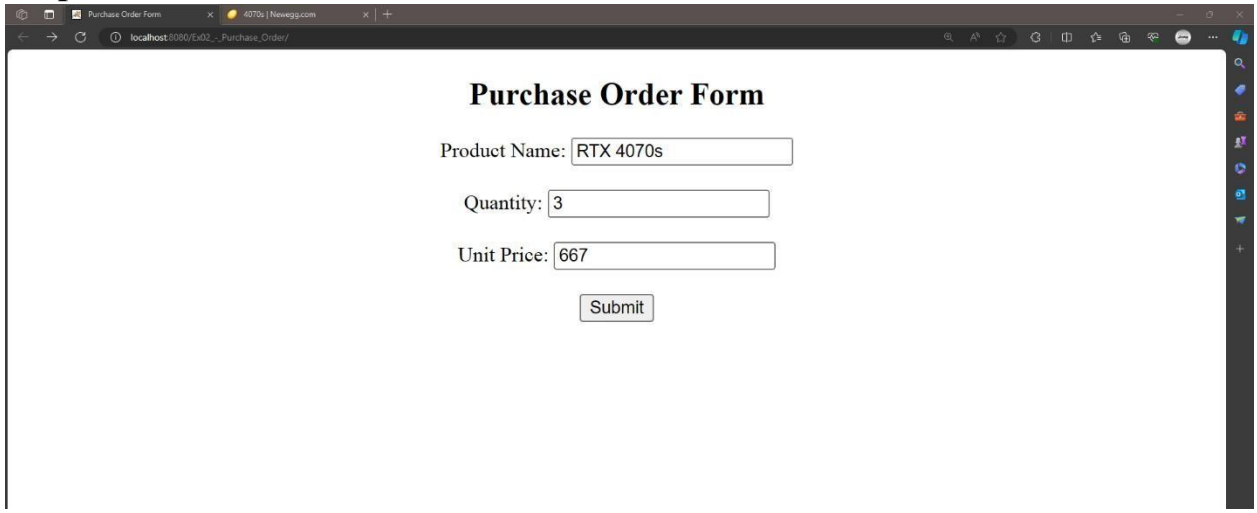
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Purchase Order Form</title>
  </head>
  <body>
    <h2>Purchase Order Form</h2>
    <form action="PurchaseOrderServlet" method="post">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" name="productName" required><br><br>
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" name="quantity" required><br><br>
      <label for="unitPrice">Unit Price:</label>
      <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
      <input type="submit" value="Submit">
    </form>
  </body>
</html>
```

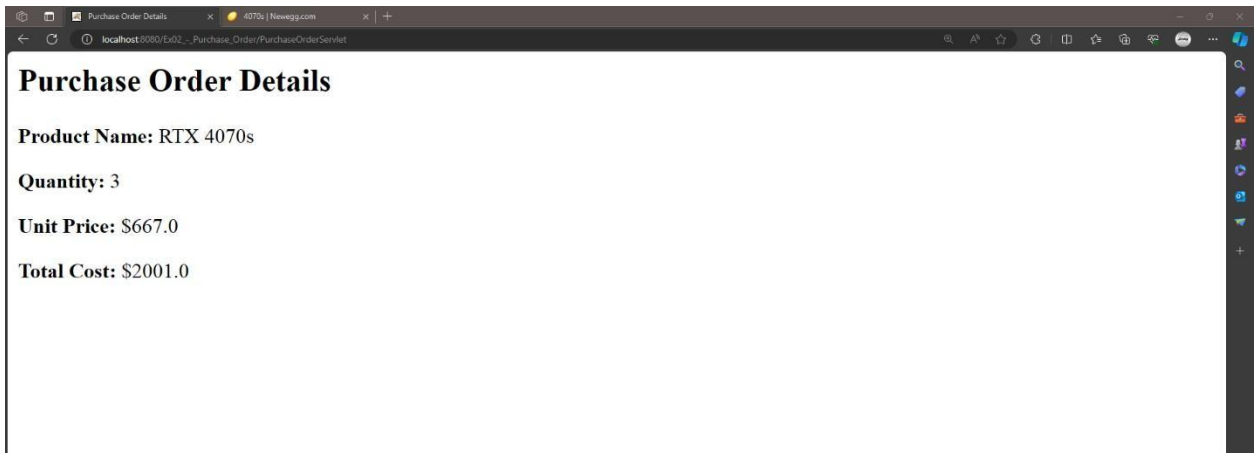
2) JAVA-Servlet(PurchaseOrderServlet.java):

```
import java.io.IOException; import java.io.PrintWriter; import
javax.servlet.annotation.WebServlet; import javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse; @WebServlet("/PurchaseOrderServlet") public
class PurchaseOrderServlet extends HttpServlet {     private static final long
serialVersionUID = 1L;     protected void doPost(HttpServletRequest request,
HttpServletResponse response)            throws IOException {
    // Retrieve form data
```

```
        String productName = request.getParameter("productName");         int
quantity = Integer.parseInt(request.getParameter("quantity"));         double unitPrice
= Double.parseDouble(request.getParameter("unitPrice"));
        // Calculate total cost         double totalCost = quantity * unitPrice;         // Display purchase
order details         response.setContentType("text/html");         PrintWriter out
= response.getWriter();         out.println("<html><head><title>Purchase Order
Details</title></head><body>");         out.println("<h2>Purchase Order Details</h2>");
out.println("<p><strong>Product Name:</strong> " + productName + "</p>");
out.println("<p><strong>Quantity:</strong> " + quantity + "</p>");
out.println("<p><strong>Unit Price:</strong> $" + unitPrice + "</p>");
out.println("<p><strong>Total Cost:</strong> $" + totalCost + "</p>");
out.println("</body></html>");
    }
}
```

Output:





RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) HTML Form (index.html):

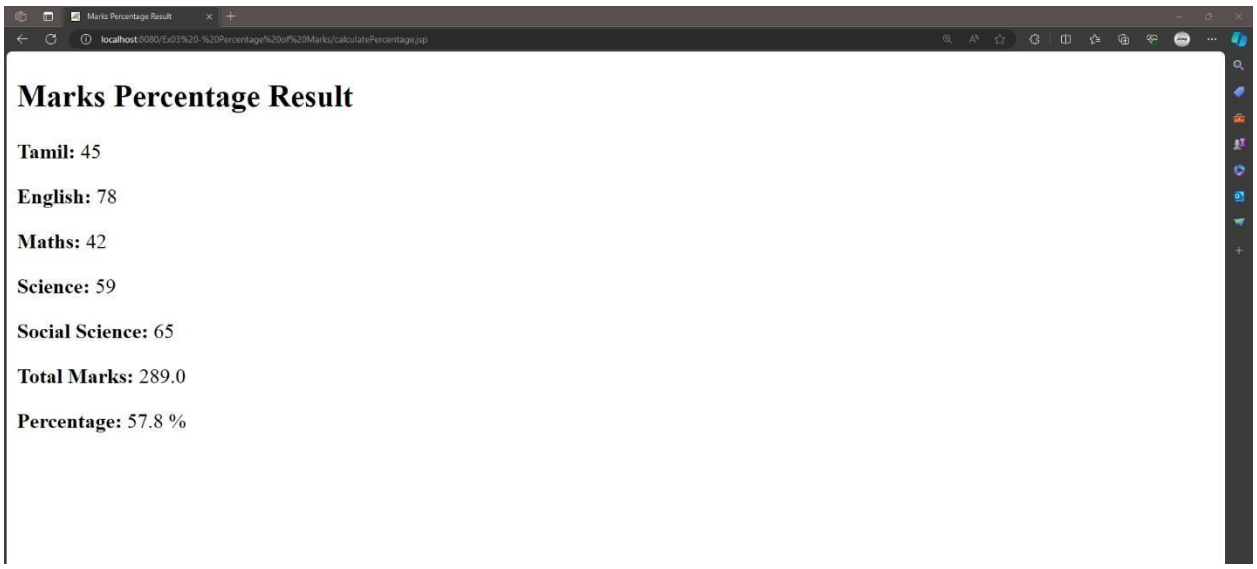
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Marks Percentage Calculator</title>
  </head>
  <body>
    <h2>Marks Percentage Calculator</h2>
    <form action="calculatePercentage.jsp" method="post">
      <label for="tamil">Tamil:</label>
      <input type="number" id="tamil" name="tamil" required><br><br>
      <label for="english">English:</label>
      <input type="number" id="english" name="english" required><br><br>
      <label for="maths">Maths:</label>
      <input type="number" id="maths" name="maths" required><br><br>
      <label for="science">Science:</label>
      <input type="number" id="science" name="science" required><br><br>
      <label for="social_science">Social Science:</label>
      <input type="number" id="social_science" name="social_science" required><br><br>
      <input type="submit" value="Calculate Percentage">
    </form>
  </body>
</html>
```

2) JSP File (calculatePercentage.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %> <% int
tamil = Integer.parseInt(request.getParameter("tamil"));     int english =
Integer.parseInt(request.getParameter("english"));     int maths =
Integer.parseInt(request.getParameter("maths"));     int science =
Integer.parseInt(request.getParameter("science"));     int social_science =
Integer.parseInt(request.getParameter("social_science"));     double totalMarks
```

```
= tamil + english + maths + science + social_science;     double percentage =
(totalMarks / 500) * 100;%>
<!DOCTYPE html>
   <head>
      <title>Marks Percentage Result</title>
   </head>
   <body>
      <h2>Marks Percentage Result</h2>
      <p> <strong>Tamil:</strong> <%= tamil%></p>
      <p> <strong>English:</strong> <%= english%></p>
      <p> <strong>Maths:</strong> <%= maths%></p>
      <p> <strong>Science:</strong> <%= science%></p>
      <p> <strong>Social Science:</strong> <%= social_science%></p>
      <p> <strong>Total Marks:</strong> <%= totalMarks%></p>
      <p> <strong>Percentage:</strong> <%= percentage%> % </p>
   </body>
</html>
```

Output:



RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) HTML File (index.html):

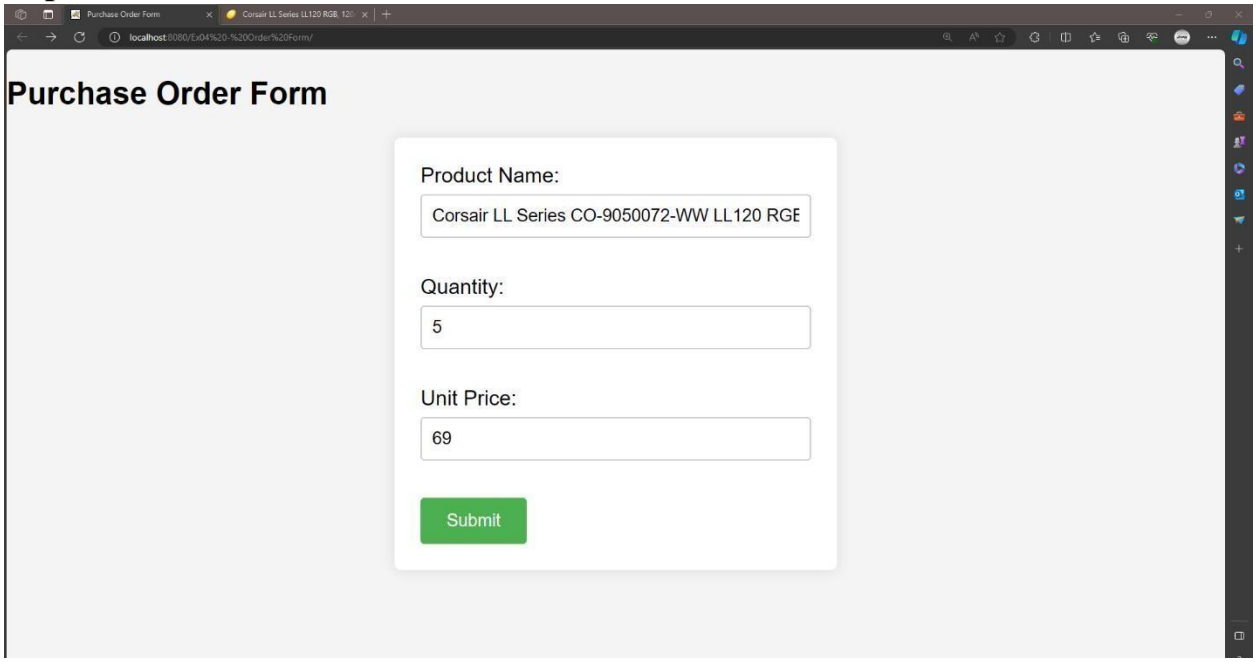
```
<!DOCTYPE html>
  <head>
    <title>Purchase Order Form</title>
  </head>
  <body>
    <h2>Purchase Order Form</h2>
    <form action="processPurchaseOrder.jsp" method="post">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" name="productName" required><br><br>
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" name="quantity" required><br><br>
      <label for="unitPrice">Unit Price:</label>
      <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
      <input type="submit" value="Submit">
    </form>
  </body>
</html>
```

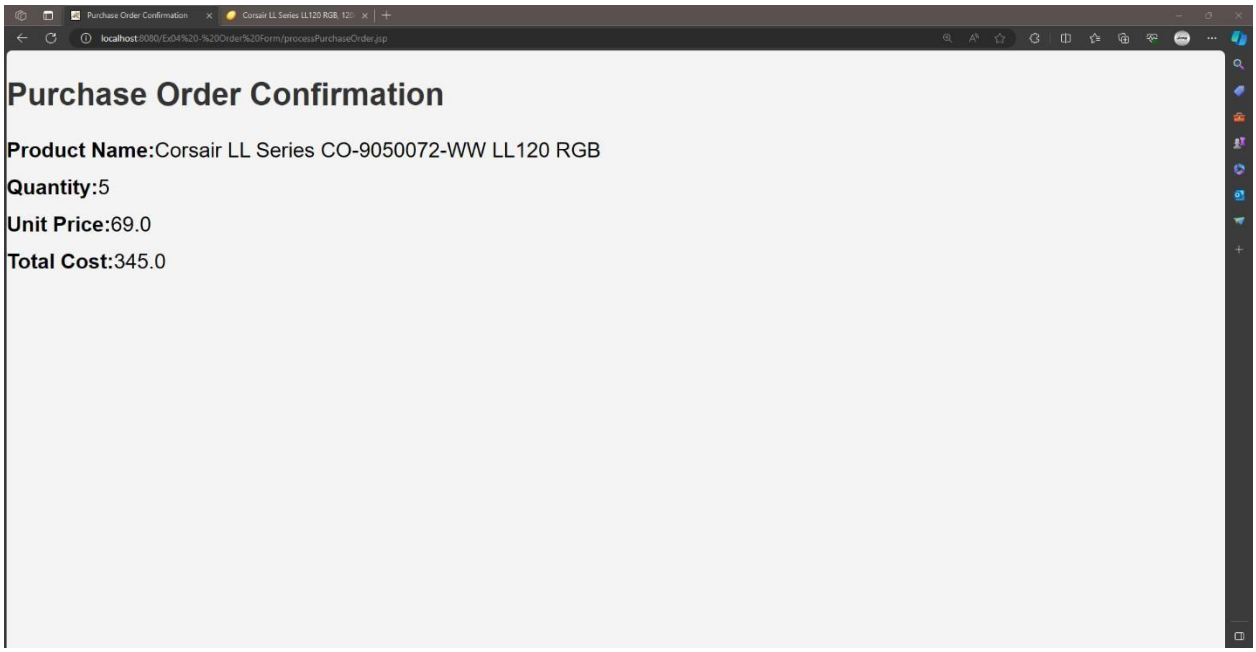
2) JSP File (processPurchaseOrder.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="java.util.*" %>
<%  String productName = request.getParameter("productName");     int
quantity = Integer.parseInt(request.getParameter("quantity"));     double
unitPrice = Double.parseDouble(request.getParameter("unitPrice"));     double
totalCost = quantity * unitPrice;%>
<!DOCTYPE html>
  <head>
    <title>Purchase Order Confirmation</title>
  </head>
  <body>
    <h2>Purchase Order Confirmation</h2>
    <p><strong>Product Name:</strong><%= productName%></p>
```

```
 <p><strong>Quantity:</strong><%= quantity%></p>
        <p><strong>Unit Price:</strong><%= unitPrice%></p>
<p><strong>Total Cost:</strong><%= totalCost%></p>
    </body>
</html>
```

Output:





RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) HTML File (index.html):

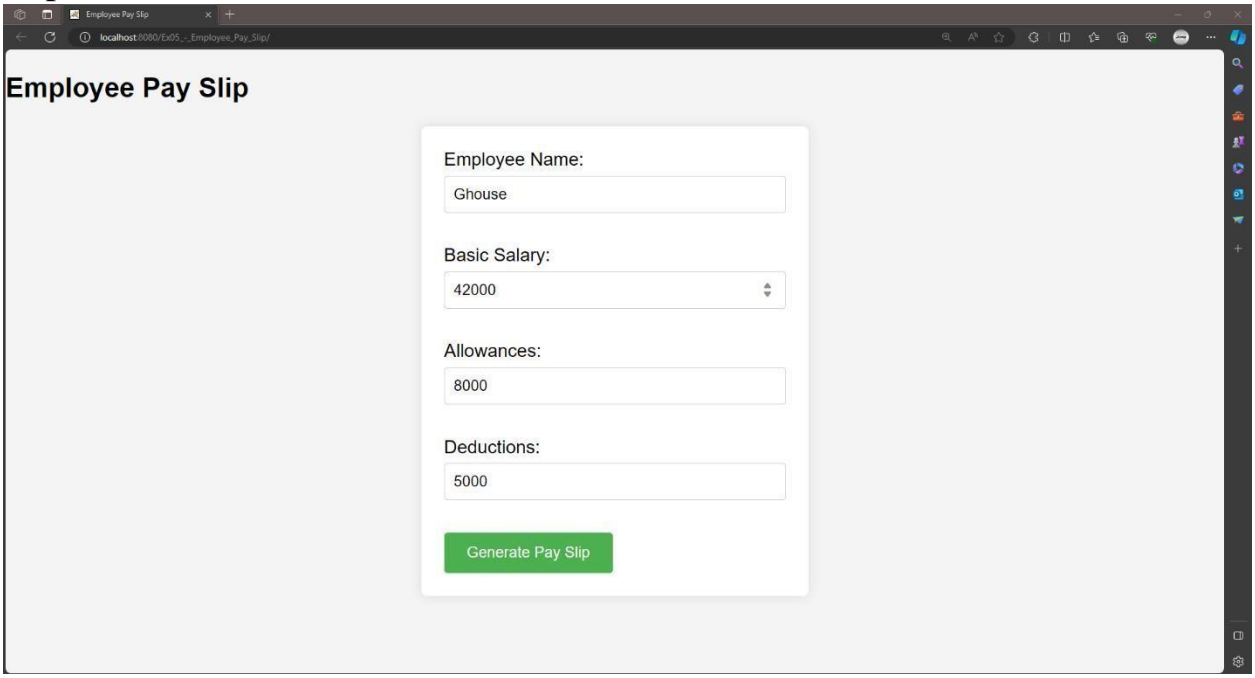
```
<!DOCTYPE html>
  <head>
    <title>Employee Pay Slip</title>
  </head>
  <body>
    <h2>Employee Pay Slip</h2>
    <form action="processPaySlip.jsp" method="post">
      <label for="employeeName">Employee Name:</label>
      <input type="text" id="employeeName" name="employeeName" required><br><br>
      <label for="basicSalary">Basic Salary:</label>
      <input type="number" id="basicSalary" name="basicSalary" required><br><br>
      <label for="allowances">Allowances:</label>
      <input type="number" id="allowances" name="allowances" required><br><br>
      <label for="deductions">Deductions:</label>
      <input type="number" id="deductions" name="deductions" required><br><br>
      <input type="submit" value="Generate Pay Slip">
    </form>
  </body>
</html>
```

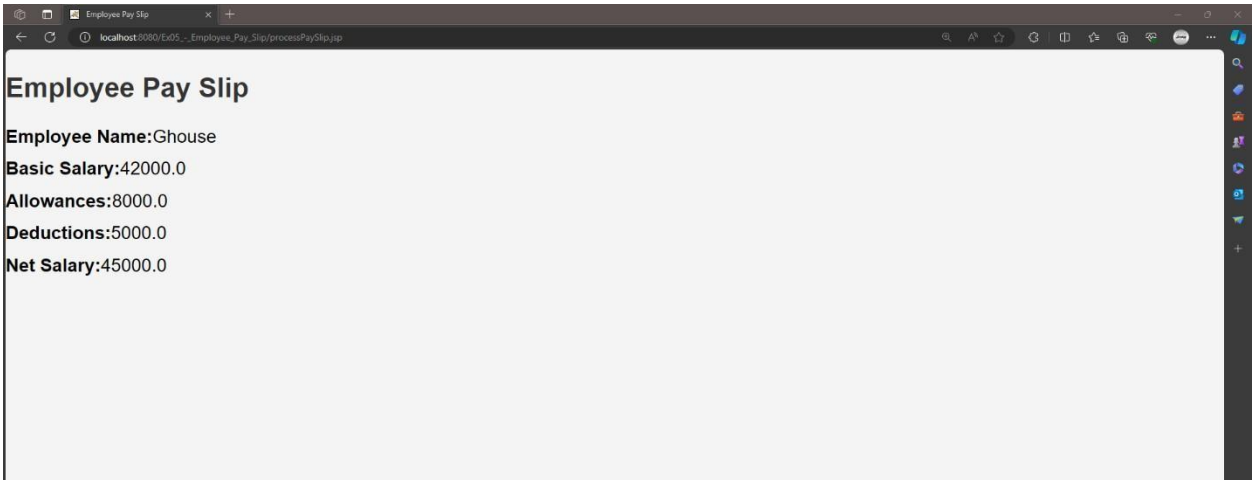
2) JSP file (processPaySlip.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%
  String employeeName = request.getParameter("employeeName");     double
basicSalary = Double.parseDouble(request.getParameter("basicSalary"));     double
allowances = Double.parseDouble(request.getParameter("allowances"));     double
deductions = Double.parseDouble(request.getParameter("deductions"));     double
netSalary = basicSalary + allowances - deductions;
%>
<!DOCTYPE html>
  <head>
    <title>Employee Pay Slip</title>
```

```
    </head>
    <body>
        <h2>Employee Pay Slip</h2>
        <p><strong>Employee Name:</strong><%= employeeName%></p>
        <p><strong>Basic Salary:</strong><%= basicSalary%></p>
        <p><strong>Allowances:</strong><%= allowances%></p>
        <p><strong>Deductions:</strong><%= deductions%></p>
        <p><strong>Net Salary:</strong><%= netSalary%></p>
    </body>
</html>
```

Output:





RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) Servlet (FormDataHandlerServlet.java):

```
import java.io.IOException; import
java.io.PrintWriter; import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(name = "FormDataHandlerServlet", urlPatterns = {"/FormDataHandlerServlet"})
public class FormDataHandlerServlet extends HttpServlet {     protected void
doPost(HttpServletRequest request, HttpServletResponse response)            throws
ServletException, IOException {        try (PrintWriter out = response.getWriter()) {
String name = request.getParameter("name");
            String email = request.getParameter("email");            String
message = request.getParameter("message");
out.println("<!DOCTYPE html>");
out.println("<html>");            out.println("<head>");
out.println("<title>Form Data</title>");
out.println("</head>");            out.println("<body>");
            out.println("<h1>Form Data</h1>");
out.println("<p>Name: " + name + "</p>");            out.println("<p>Email:
" + email + "</p>");            out.println("<p>Message: " + message +
"</p>");            out.println("</body>");            out.println("</html>");
        }
    }
}
```

2) HTML File (index.html):

```
<!DOCTYPE html>
<html>
    <head>
        <title>Form Submission</title>
    </head>
```

```
    <body>
        <h1>Submit Form</h1>
        <form action="FormDataHandlerServlet" method="post">
            <label for="name">Name:</label><br>
            <input type="text" id="name" name="name"><br>
            <label for="email">Email:</label><br>
            <input type="email" id="email" name="email"><br>
            <label for="message">Message:</label><br>
            <textarea id="message" name="message"></textarea><br><br>
            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

Output:

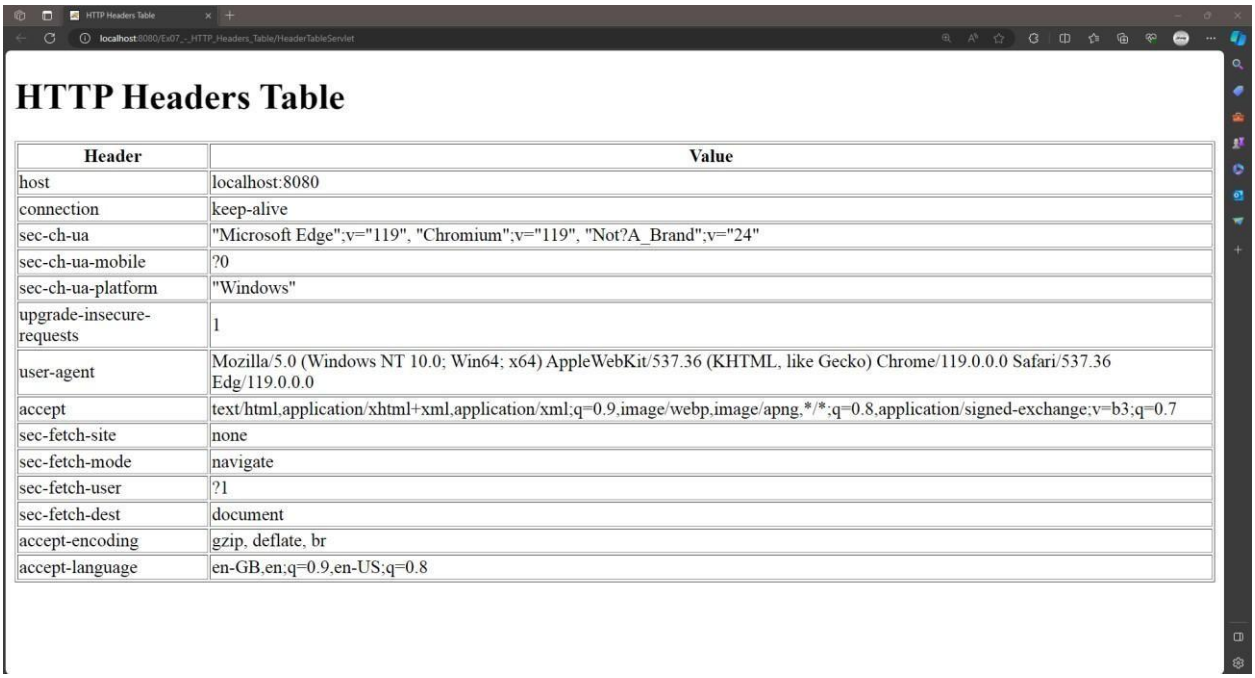
RESULT:

Thus the above program has been created and executed successfully

Source Code:

```
import java.io.IOException; import
java.io.PrintWriter; import java.util.Collections;
import java.util.Enumeration; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(name = "HeaderTableServlet", urlPatterns = {"/HeaderTableServlet"})
public class HeaderTableServlet extends HttpServlet {     protected void
doGet(HttpServletRequest request, HttpServletResponse response)            throws
ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");        try (PrintWriter out =
response.getWriter()) {            out.println("<!DOCTYPE html>");            out.println("<html>");
out.println("<head>");
out.println("<title>HTTP Headers Table</title>");            out.println("</head>");
out.println("<body>");                out.println("<h1>HTTP Headers Table</h1>");
out.println("<table border='1'>");
        out.println("<tr><th>Header</th><th>Value</th></tr>");
Enumeration<String> headerNames = request.getHeaderNames();            while
(headerNames.hasMoreElements()) {
            String headerName = headerNames.nextElement();
            Enumeration<String> headerValues = request.getHeaders(headerName);
while (headerValues.hasMoreElements()) {
                String headerValue = headerValues.nextElement();
out.println("<tr><td>" + headerName + "</td><td>" + headerValue + "</td></tr>");                }
        }
        out.println("</table>");
out.println("</body>");            out.println("</html>");
    }
  }
}
```

Output:



HTTP Headers Table

Header	Value
host	localhost:8080
connection	keep-alive
sec-ch-ua	"Microsoft Edge";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile	?0
sec-ch-ua-platform	"Windows"
upgrade-insecure-requests	1
user-agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36 Edg/119.0.0.0
accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
sec-fetch-site	none
sec-fetch-mode	navigate
sec-fetch-user	?1
sec-fetch-dest	document
accept-encoding	gzip, deflate, br
accept-language	en-GB,en;q=0.9,en-US;q=0.8

RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) Remote Interface: (CollegeInterface.java):

```
import java.rmi.Remote; import
java.rmi.RemoteException; public interface CollegeInterface
extends Remote {
   String getStudentInfo(int studentId) throws RemoteException;
}
```

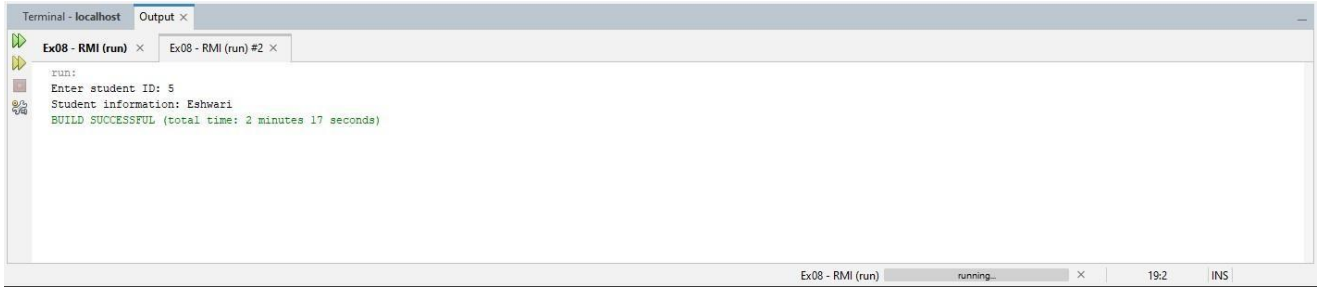
2) Server Class: (CollegeServer.java): import

```
java.rmi.RemoteException; import java.rmi.registry.LocateRegistry; import
java.rmi.registry.Registry; import
java.rmi.server.UnicastRemoteObject; import java.util.HashMap; import
java.util.Map;
public class CollegeServer extends UnicastRemoteObject implements
CollegeInterface {     private Map<Integer, String> studentRecords;
public CollegeServer() throws RemoteException {         super();
       studentRecords = new HashMap<>();
studentRecords.put(5, "Eshwari");         studentRecords.put(8,
"Kalaiarasi");         studentRecords.put(20, "Savitha");
   }
   public String getStudentInfo(int studentId) throws RemoteException {
return studentRecords.getOrDefault(studentId, "Student not found");
   }
   public static void main(String[] args) {
try {
          CollegeServer obj = new CollegeServer();
          Registry registry = LocateRegistry.createRegistry(1099);
registry.rebind("CollegeServer", obj);
          System.out.println("Server ready");
       } catch (Exception e) {
          System.err.println("Server exception: " + e.toString());
e.printStackTrace();
       }
   }
}
```

3) Client Class: (CollegeClient.java)):

```
import java.rmi.registry.LocateRegistry; import
java.rmi.registry.Registry; import
java.util.Scanner; public class CollegeClient {
public static void main(String[] args) {        try
{
        Registry registry = LocateRegistry.getRegistry("localhost", 1099);
        CollegeInterface stub = (CollegeInterface) registry.lookup("CollegeServer");
        Scanner scanner = new Scanner(System.in);
System.out.print("Enter student ID: ");            int studentId
= scanner.nextInt();
        String studentInfo = stub.getStudentInfo(studentId);
        System.out.println("Student information: " + studentInfo);
    } catch (Exception e) {
        System.err.println("Client exception: " + e.toString());
e. printStackTrace();
    }
  }
}
```

Output:



RESULT:

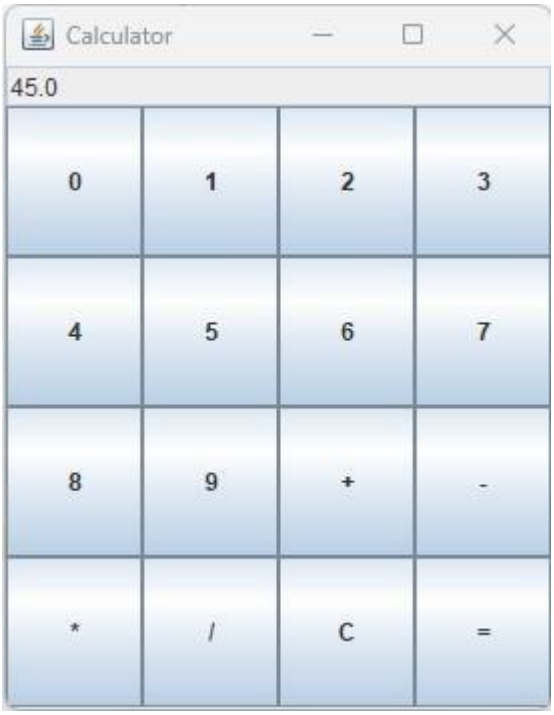
Thus the above program has been created and executed successfully

Source Code:

```
import java.awt.*; import java.awt.event.*; import javax.swing.*;
public class Calculator extends JApplet implements ActionListener {
private JTextField textField;      private JButton[] numberButtons;
private JButton[] operatorButtons;      private JButton clearButton;
private JButton equalsButton;      private String displayText = "";
private double result = 0;    private String operator = "";    public void
init() {          setLayout(new BorderLayout());          textField = new
JTextField();          textField.setEditable(false);          add(textField,
BorderLayout.NORTH);          JPanel buttonPanel = new JPanel();
buttonPanel.setLayout(new GridLayout(4, 4));       numberButtons =
new JButton[10];                    for (int i = 0; i < 10; i++) {
numberButtons[i]       =       new       JButton(String.valueOf(i));
numberButtons[i].addActionListener(this);
buttonPanel.add(numberButtons[i]);
      }
      operatorButtons = new JButton[4];
operatorButtons[0] = new JButton("+");
operatorButtons[1] = new JButton("-");       operatorButtons[2]
= new JButton("*");       operatorButtons[3] = new
JButton("/");       for
(int i = 0; i < 4; i++) {
operatorButtons[i].addActionListener(this);          buttonPanel.add(operatorButtons[i]);
      }
      clearButton = new JButton("C");
clearButton.addActionListener(this);       buttonPanel.add(clearButton);
equalsButton
= new JButton("=");
equalsButton.addActionListener(this);
buttonPanel.add(equalsButton);
add(buttonPanel, BorderLayout.CENTER);
   }
   public void actionPerformed(ActionEvent e) {
JButton source = (JButton) e.getSource();
```

```
String buttonText = source.getText();        if
(buttonText.equals("C")) {            displayText =
"";            result = 0;            operator = "";
        } else if (buttonText.equals("=")) {            double
secondOperand = Double.parseDouble(displayText);            switch
(operator) {                case "+":                    result += secondOperand;
break;                case "-":
                result -= secondOperand;
break;                case "*":
                result *= secondOperand;
break;                case "/":
                if (secondOperand != 0) {
result /= secondOperand;                    } else
{                    displayText = "Error:
Division by zero";
                }
break;
            }
            displayText = String.valueOf(result);
operator = "";
        } else if (buttonText.equals("+") || buttonText.equals("-")
|| buttonText.equals("*") || buttonText.equals("/")) {            operator
= buttonText;            result =
Double.parseDouble(displayText);            displayText = "";
        } else {            displayText
+= buttonText;
        }
        textField.setText(displayText);
    }
}
```

Output:



RESULT:

Thus the above program has been created and executed successfully

Source Code:

1) Receiver Project (Main.java):

```
import
java.io.DataOutputStream; import
java.net.Socket; public class Main {     public
static void main(String[] args) {        try {
        String messageToSend = "Que Ota? (Translation: Want another one?)";
        Socket socket = new Socket("127.0.0.1", 12345);
        DataOutputStream out = new DataOutputStream(socket.getOutputStream());
outwrite(Massagetae);
        out. flush();
out.close();            socket.close();
        System.out.println("Message sent successfully.");
} catch (Exception e) {            e.printStackTrace();
      }
   }
}
```

2) Sender Project (Main.java):

```
import java.io.DataInputStream; import
java.net.ServerSocket;                import
java.net.Socket; public class Main {
public static void main(String[] args) {
try {
        ServerSocket serverSocket = new ServerSocket(12345);
        System.out.println("Waiting for incoming connection...");
        Socket socket = serverSocket.accept();
        DataInputStream in = new DataInputStream(socket.getInputStream());
        String receivedMessage = in.readUTF();
        System.out.println("Received message: " + receivedMessage);
in.close();            socket.close();            serverSocket.close();        }
catch (Exception e) {            e.printStackTrace();
      }
```

```
    }
}
```

Output:

RESULT:

Thus the above program has been created and executed successfully